

Neuro-Symbolic AI in 2024: A Systematic Review

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Abstract

Background: The field of Artificial Intelligence has undergone cyclical periods of growth and decline, known as AI summers and winters. Currently, we are in the third AI summer, characterized by significant advancements and commercialization, particularly in the integration of Symbolic AI and Sub-Symbolic AI, leading to the emergence of Neuro-Symbolic AI.

Contributions: (1) A definition of Meta-Cognition within Neuro-Symbolic AI. (2) A review of the key themes of the literature post the Neuro-Symbolic research explosion from 2020-2024. (3) Identification of the current gaps in the literature of Neuro-Symbolic AI

Objective: This paper provides a systematic literature review of Neuro-Symbolic AI projects within the 2020-24 AI landscape, highlighting key developments, methodologies, and applications. It aims to identify where quality efforts are focused in 2024 and pinpoint existing research gaps in the field.

Methods: The review followed the PRISMA methodology, utilizing databases such as IEEE Explore, Google Scholar, arXiv, ACM, and SpringerLink. The inclusion criteria targeted peer-reviewed papers published between 2020 and 2024. Papers were screened for relevance to Neuro-Symbolic AI, with further inclusion based on the availability of associated codebases to ensure reproducibility.

Results: From an initial pool of 1,428 papers, 167 met the inclusion criteria and were analyzed in detail. The majority of research efforts are concentrated in the areas of learning and inference (63%), logic and reasoning (35%), and knowledge representation (44%). Explainability and trustworthiness are less represented (28%), with Meta-Cognition being the least explored area (5%). The review identifies significant interdisciplinary opportunities, particularly in integrating explainability and trustworthiness with other research areas.

Discussion: The findings reveal a well-integrated body of work in learning and inference, logic and reasoning, and knowledge representation. However, there is a notable gap in research focused on explainability and trustworthiness, which is critical for the deployment of reliable AI systems. The sparse representation of Meta-Cognition highlights the need for further research to develop frameworks that enable AI systems to self-monitor, evaluate, and adjust their processes, enhancing autonomy and adaptability.

Conclusion: Neuro-Symbolic AI research has seen rapid growth since 2020, with concentrated efforts in learning and inference. Significant gaps remain in explainability, trustworthiness, and Meta-Cognition. Addressing these gaps through interdisciplinary research will be crucial for advancing the field towards more intelligent, reliable, and context-aware AI systems.

Keywords

Neuro-Symbolic AI, Systematic Review, Learning and Inference, Knowledge Representation, Logic and Reasoning, Explainability and Trustworthiness, Meta-Cognition, PRISMA.

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1. Introduction

The field of Artificial Intelligence (AI) has experienced significant cyclical growth, known as AI summers and winters. At present, we as a community find ourselves in the third AI summer, marked by rapid scientific advances and commercialization, continuing the legacy of previous periods of AI excitement followed by setbacks [1]. A significant product of the third AI summer has been the integration of two prominent fields of AI; Symbolic AI and Sub-Symbolic AI, the fusion of which is known as Neuro-Symbolic AI. There is an ongoing debate about the necessity of Neuro-Symbolic AI [2], opponents arguing that common sense reasoning can be addressed through the use of big data [3] and proponents arguing that “You can’t get to the moon by climbing successively taller trees” [4]. For this systematic review, we take the stance that symbolic AI is essential and that Neuro-Symbolic AI represents the best way forward for the community hence, this paper provides a systematic literature review of prominent Neuro-Symbolic projects within the 2024 AI landscape, highlighting key developments, methodologies, and applications.

1.1. Symbolic AI

Symbolic AI is a “a sub-field of AI concerned with learning the internal symbolic representations of the world around it” where we can “translate some form of implicit human knowledge into a more formalized and declarative form based on rules and logic” [2]. Examples of some of the earliest AI systems that utilised symbolic representations include SHRDLU [5], ELIZA [6], DENDRAL [7] and MYCIN[8] and examples of some of the newest AI systems which heavily utilise symbolic processes include ConceptNet 5.5 [9] CYC [10] and Good Old Fashioned AI (GOFAI) planning systems [11] to name just a few.

1.2. Sub-Symbolic AI

By contrast, Sub-Symbolic AI are systems that “do not require rules or symbolic representations as inputs” and instead “learn implicit data representations on their own” [2]. Sub-Symbolic AI encompasses approaches such as machine learning, deep learning, and generative AI, which rely on algorithms to automatically extract patterns from raw data to discern relationships and make predictions based on learned representations. Examples of some of the earliest AI systems that utilised sub-symbolic representations include the Perceptron [12], Hopfield Networks [13] and the Backpropagation Algorithm [14] and examples of some of the newest sub-symbolic systems include famous projects such as the Generative Pre-trained Transformer (GPT) models [15], the YOLO family of Convolutional Neural Networks (CNN’S) [16] and the DALLE diffusion model transformer [17] to again just name a few.

1.3. Neuro-Symbolic AI

There is at present a debate within the AI community surrounding the need for Neuro-Symbolic AI [18]. Simply described, the argument for Neuro-Symbolic AI draws on Kahneman (2011) [19] concepts of System 1 and System 2 thinking whereby System 1 is fast, intuitive, and parallel, akin to the capabilities of deep learning, while System 2 is slow, deliberate, and sequential,

resembling symbolic reasoning and hence, Neuro-Symbolic AI aims to combine these two approaches to create systems that benefit from the strengths of both. We Adopt the definition provided by Garcez and Lamb (2023) [18]; Hence, Neuro-Symbolic AI is “**a composite AI framework that seeks to merge the domains of Symbolic AI and Neural Networks**” [or more broadly put, Sub-Symbolic AI] “**to create a superior hybrid AI model possessing reasoning capabilities**”. As this definition is quite broad, for the purpose of this systematic review, we will further define the sub-components of the Neuro-Symbolic AI taxonomy we believe to be most relevant to the current AI landscape within section 2.

2. Methodology

2.1. Taxonomy of Neuro-Symbolic AI

We identified five foundational research areas advancing the state of the art in Neuro-Symbolic AI. This taxonomy was synthesized from a review of six survey papers [20, 21, 22, 23, 24, 25] and four seminal books [2, 26, 27, 28]. These areas are:

1. **Knowledge Representation:** Integrating symbolic and neural representations and developing commonsense and domain-specific knowledge graphs [20, 26, 28].
2. **Learning and Inference:** Combining learning and reasoning processes through end-to-end differentiable reasoning and dynamic multi-source knowledge reasoning [21, 22, 2].
3. **Explainability and Trustworthiness:** Creating interpretable models and reasoning processes to ensure trust and reliability in Neuro-Symbolic systems [23, 24].
4. **Logic and Reasoning:** Integrating logic-based methods with neural networks, including logical and probabilistic reasoning, and the syntax and semantics of Neuro-Symbolic systems [23, 28].
5. **Meta-Cognition:** The system’s capacity to monitor, evaluate, and adjust its own reasoning and learning processes by integrating neural networks and symbolic representations.

The four above categories represent the core technical areas where current efforts are concentrated. Additionally, we define *Meta-Cognition* to address a gap in current taxonomies that fail to capture fields encompassing self-awareness, adaptive learning, reflective reasoning, self-regulation, and introspective monitoring.

2.2. Meta-Cognition

Meta-Cognition refers to the processes that involve thinking about one’s thinking, enabling self-awareness and self-regulation in cognitive tasks. Meta-Cognition is the controller that sits above the cognitive tasks of systems to direct energy effectively towards the correct system to handle a task. This higher-order cognition is crucial for tasks that require reflection, planning, and adaptation. Its importance lies in its ability to enhance learning, problem-solving, and